

# Experimental study of catalytic pyrolysis of polyethylene and polypropylene over USY zeolite and separation to gasoline and diesel-like fuels



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## ABSTRACT

Thermal catalytic and non catalytic degradation of polyethylene (PE) and polypropylene (PP) were carried out in a batch reactor at 450 °C. The major product obtained from the non-catalytic pyrolysis of PE was 80 wt% of wax, while the degradation of PP without any catalyst produced 85.5% of liquid. In catalytic degradation, ultra-stable Y (USY) zeolite was used in a ratio of 1:10, with respect to PE and PP. In both cases, polymers were converted to liquid products with high yields (71 and 82 wt% respectively) and a low amount of coke deposit has appeared. The liquid fraction derived from the catalytic pyrolysis of PE was a mixture of C<sub>5</sub>–C<sub>30</sub> compounds and that of PP was a mixture of C<sub>5</sub>–C<sub>30</sub>. This wide range of components suggests the separation of liquids into two phases in order to use them as diesel-like and gasoline-like fuels. Three temperatures of separation (130, 150, 170 °C) were tested and the physical properties were compared. Among the tested temperatures, 170 °C was the optimal temperature of separation where the distillation curves and physical properties of the light and heavy phases fitted completely those of gasoline and diesel fuels, respectively. After separation, the gasoline-like fuel which represents 60.6% of the total oil of PP and 57% of that of PE had high octane numbers (RON = 96 and RON = 97, respectively), while the diesel-like fuel which accounts for 36.5% of the total oil of PP and 35.3% of that of PE have high calculated cetane numbers (52 and 53, respectively).

## 1. Introduction

Due to rapid urbanization and economic development, the world's annual production of plastic materials has increased from around 1.7 million tons in the 1950s to nearly 311 million tons, of which 59 million tons are accounted for by Europe alone [1]. Due to the increase in generation, waste plastics are becoming a major stream in solid wastes, generating heavy environmental problems.

There are six main plastic components in European municipal solid waste which are high-density polyethylene (HDPE), low density polyethylene (LDPE), polypropylene (PP), polystyrene (PS), polyvinylchloride (PVC) and polyethyleneterephthalate (PET). Polyolefin materials (PE and PP) constitute the majority of plastic wastes and represent around 68% of the municipal solid wastes [2].

A major part of post consumer plastic wastes are currently landfilled or incinerated. Since these wastes are non biodegradable and the areas for landfill sites are limited, landfilling is not a suitable option for disposing plastic wastes. In addition, the use of incinerator have high cost consumption, and generates some pollutants to the air like nitrous and sulfur oxides, dusts, dioxins and furan [3]. Therefore, in order to

reduce environmental charge, recycling and recovering methods should be employed. Pyrolysis is a recovery method that consists on upgrading plastic wastes into fuel oil and valuable chemicals by heating at temperature range of 400–600 °C in the absence of oxygen [4].

Thermal and catalytic degradation of waste plastics are two kinds of chemical valorization processes. The presence of catalysts in pyrolysis reduces reaction temperature and residence time, decreases energy consumption, promotes decomposition reaction and improves selectivity and the quality of the products [5].

Pyrolysis of waste polyethylene has been investigated by many researchers who found that waxes are the main products in thermal cracking. Waxes derived from thermal cracking of PE are long chain made up of linear hydrocarbons having a carbon number heavier than C<sub>20</sub> and which solidify at room temperature [6]. Kiran et al. [7] studied the pyrolysis of waste polyethylene in a fixed bed reactor, at a low heating rate and reaching 600 °C. Gas and wax were the two main products obtained during the pyrolysis experiments. The wax formed caused an operating problem by plugging the product lines and condenser tubes.

In order to improve the quality of the liquid fuel produced from the

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degradation of plastics, a range of catalysts have been utilized in the catalytic degradation of polymers. The most commonly used catalysts are solid acid ones such as zeolites and silica–alumina [8]. Lerici et al. [9] studied thermocatalytic degradation of polyethylene, polypropylene (PP) and polystyrene (PS) using the H-Y zeolite in a batch reactor at 500 °C. They reported that thermal cracking of polyolefins produced waxes and the use of catalyst yielded higher percentages of gaseous products which ranged between 42 wt% and 45 wt%, while PS yielded a greater amount of liquids (~72 wt%).

A number of authors have reported promising results on the cracking of polyolefins over several catalysts. Onu et al. [10] compared the degradation of polyethylene and polypropylene over HZSM-5 zeolite catalyst and a catalyst modified with orthophosphoric acid (PZSM-5) in a fixed bed reactor at 460–480 °C. According to their study, the quantities of liquid products with PZSM-5 are smaller than those obtained over HZSM-5 for both PP (43.3%) and PE (34.6%), and all of the liquids had high proportions of aromatic hydrocarbons.

The cracking ability of a catalyst is related to the acidity strength and the pore size which represent its main characteristics [11]. Strong acid catalysts enhance the cracking of heavier hydrocarbons into more lighter or gaseous hydrocarbons comparing to the weak acid catalysts. This was reported by Sakata et al. [12] who studied the catalytic degradation of HDPE into fuel oil over mesoporous silica (KFS-16) catalyst at 430 °C. The Product yields and composition were compared with those obtained over solid acid catalyst (silica–alumina and ZSM-5). According to their study, ZSM-5 possessing strong acid sites produced less liquid products and more gaseous products than the silica–alumina catalysts. On the other hand, KFS-16, having no acid sites, produced liquid hydrocarbons with a high yield as 71 wt%.

Sakata et al. [13] also investigated the effects of various types of solid acid catalysts such as silica–alumina, zeolite and non-acidic mesoporous silica catalysts on the degradation of polyethylene and polypropylene at 430 and 380 °C respectively in batch operation. The degradation of PP and PE into gases was accelerated when using catalysts with strong acid sites such as zeolite ZSM-5.

A number of authors have investigated the use of USY zeolite in the catalytic degradation of polyethylene and polypropylene [2,14,15]. They reported that large-pore USY zeolite produced the highest amount of liquid fraction compared to other tested catalysts like ZSM-5 zeolite which favored gas production. Moreover, the hydrocarbons formed over USY zeolite were heavier than those formed with medium-pore zeolites and the main products were alkanes with less alkenes and aromatics.

The main disadvantage of using zeolite catalyst is coke formation which is due to the deposit of heavy by-products on the catalyst surface. Coke formation occurs during catalytic reactions and causes the gradual deactivation of the catalyst which also affects the product distribution and selectivity [16]. Several authors investigated the deactivation of zeolites in polymer cracking.

Marcilla et al. [17] studied the deactivation of HZSM5 and HUSY zeolites during the catalytic pyrolysis of polyethylene and reported that HUSY showed a higher initial relative activity as it presents large pore size but it lead to a fast deactivation, while small pore size in HZSM5 prevented coke deposition.

Previous studies have compared the thermal and catalytic degradation of polyethylene and polypropylene, but most of these works were performed using catalysts that favored gas production [18] and did not focused on the quality of the liquid fraction produced. In the present work, the main objectives were to produce useful fuels from PE and PP over USY zeolite which enhance liquid formation and to investigate the quality of the liquid fraction produced. Furthermore, the influence of USY zeolite on the yield and composition of the derived oils from catalytic cracking of PP and PE was studied. Moreover, the separation of the pyrolytic oils of PP and PE into light, gasoline-like and heavy, diesel-like fractions was studied in order to find a single temperature where the boiling range distributions and physical properties

**Table 1**

Textural and acid properties of the USY zeolite.

|                                                            |         |
|------------------------------------------------------------|---------|
| SiO <sub>2</sub> /Al <sub>2</sub> O <sub>3</sub> (mol/mol) | 15      |
| BET surface area (m <sup>2</sup> g <sup>-1</sup> )         | 550     |
| Na <sub>2</sub> O (wt%)                                    | 0.05    |
| Micropore volume (MPV) (cm <sup>3</sup> g <sup>-1</sup> )  | 0.1     |
| Crystal size (μm)                                          | 0.7–1.0 |
| Total acidity (mmol NH <sub>3</sub> g <sup>-1</sup> )      | 0.1     |

of each fraction meet the gasoline and diesel fuels standards.

## 2. Experimental

### 2.1. Materials and methods

The feedstock samples for the present study were virgin plastic mainly made of PE and PP plastic types respectively thus the material variability can be eliminated. All of these plastics were granular with a particle size of, approximately, 1.8 mm–4.5 mm.

The catalyst used for the pyrolysis experiments was a commercial USY zeolite provided by TOSOH Corporation. Their main characteristics are presented in table 1. This zeolite has been selected for its medium Si/Al ratio (7.5). Indeed, for low Si/Al ratios, the number of acid sites will increase, but their strength remains relatively less pronounced. However, a higher Si/Al ratio induces a drastic decrease of the number of acid sites. In this case, “isolated” acid sites will be more reactive. Hence, the selected Si/Al ratio seems to be a good compromise in order to enhance liquid production at the expense of gases [19].

The catalyst was dried in an oven to remove moisture, and then was sieved in order to have a particle size between 200 and 800 μm. In the catalytic pyrolysis experiments the plastic/catalyst mass ratio was 10/1.

### 2.2. Laboratory scale pyrolysis plant

Pyrolysis reactions were led in a laboratory scale batch reactor, developed by Pyrum Innovations (Fig. 1). The reactor is 200 mm height with 90 mm diameter made of steel and heated by an electric coil with a power of 1100 W, which is connected to a programmable temperature controller. The reactor is thermally insulated with ceramic wool jacket. A thermocouple in direct contact with the sample was used to monitor the actual temperature of the process.

Prior to the experiment, the system was purged for 15 min with nitrogen (N<sub>2</sub>) at room temperature to ensure an inert atmosphere, and then 2 series of experiments were carried out. In the first series, 100 g of PP and PE were loaded individually without any catalyst into the reactor, while in the second series; 100 g of PP and PE were mixed individually with 10 g of catalyst and loaded into the reactor. The temperature of the reactor was then raised from ambient temperature to 450 °C. The pyrolysis products pass through a stainless steel condenser cooled with tap water at 15 °C and then, the condensate is collected in a container. The uncondensed gases are discharged through an exhaust system. The products of degradation were classified into three groups: liquid products, gaseous products and residues. The liquid fraction was then characterized in order to be used as an alternative fuel oil. For this purpose liquid product was fractionated at different temperatures to find the optimal temperature for separating the liquid into gasoline-like and diesel-like oil fractions.

### 2.3. Liquid products characterization

The samples were analyzed to determine the main parameters. The calorific value of the samples was measured using a Parr 6200 calorimeter. The density of the pyrolytic oil was measured at room

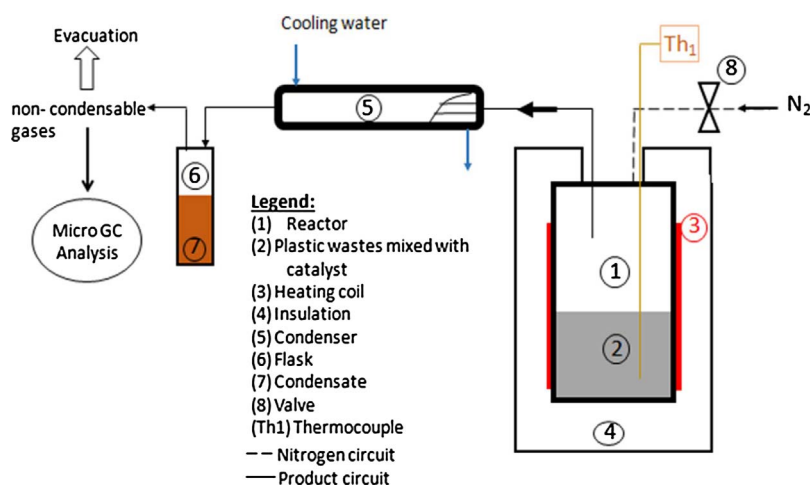


Fig. 1. Schematic diagram of pyrolysis plant.

temperature using pycnometer M50T (700–750 G/L) and a pycnometer M50T (750–800 G/L). The viscosity of the oils produced in each experiment were measured at 40 °C using an AND vibro viscometer, and the distillation curve of the obtained oil was measured by a NORMALAB NDI440 automatic distiller. The vapor pressure of the light fraction was measured by a vapor pressure analyzer ERAVAP using ASTM D6378 testing method, while the flash point of the heavy phase was measured using a PENSKEY MARTENS NPM440 device with a precision of 2 °C. The analyses were realized in three replicates, and error bars were used to represent the standard error.

The cetane index (CCI) was calculated using temperature values of 10%, 50% and 90% of recovered volume ( $T_{10}$ ,  $T_{50}$  and  $T_{90}$ ) and of density at 15 °C (D) in an equation of four variables [20] presented in Eq. (1).

$$CCI = 45.2 + 0.0892T_{10N} + (0.131 + 0.901B)T_{50N} + (0.0523 - 0.420B)T_{90N} + 0.00049(T_{10N}^2 - T_{90N}^2) + 107B + 60B^2 \quad (1)$$

Where  $T_{10N} = T_{10} - 215$ ,  $T_{50N} = T_{50} - 260$ ,  $T_{90N} = T_{90} - 310$   
 $And B = (e^{-3.5(D-0.85)}) - 1$

The octane number was calculated using a correlation between the research octane number (RON) of unleaded motor gasoline and its 90% volume boiling point [21] shown in Eq. (2).

$$RON = 179.5415 - 0.136428T_{90\%} - 1.3072 \times 10^{-3}T_{90\%}^2 - 113222.4 \times 10^3 T_{90\%}^3 \quad (2)$$

In order to quantify the dominant chemical compounds of the produced oils, samples were analyzed using a Perkin Elmer CLARUS 600 Mass Spectrometer coupled with a gas chromatograph and a flame ionization detector (GCMS-FID). An Agilent SLB-5MS column (30 m × 0.250 mm; 0.25 μm film thickness) was used with a helium flow rate of 1 ml/min. The temperature program began with a hold at 70 °C for 1 min followed by an increase at 12 °C/min to 180 °C, then an increase at 7 °C/min to 240 °C and finally an increase at 12 °C/min to 330 °C, which was held for 6.76 min. The temperature of FID detector was fixed at 275 °C. Injections of 0.5 μl of oil samples were realized with a split ratio of 10:1. The mass spectrometer was set at an ionizing voltage of 70 eV and a range of  $m/z$  30–450. Identification of chromatographic peaks was established from MS data.

### 3. Results and discussion

#### 3.1. Pyrolysis yields (wt%)

The yields of liquid, gaseous and solid products obtained at 450 °C from the non-catalytic and catalytic degradation were calculated using Eq. (3), and results are presented in Table 2.

**Table 2**  
Effect of catalyst type on the product yields and reaction time

| Plastic type        | Non-catalytic Degradation |      | Catalytic Degradation |      |
|---------------------|---------------------------|------|-----------------------|------|
|                     | PE                        | PP   | PE                    | PP   |
| Yield (wt.%)        |                           |      |                       |      |
| Wax                 | 80                        | –    | –                     | –    |
| Liquid              | –                         | 85.5 | 71                    | 82   |
| Solid               | –                         | –    | 2.2                   | 1.2  |
| Gas                 | 20                        | 14.5 | 26.8                  | 16.8 |
| Reaction time (min) | 120                       | 80   | 50                    | 45   |

$$Yield(\%) = \frac{m_i}{m} \quad (3)$$

Where  $m_i$  is the weight of products (liquid oil, wax and residue) and  $m$  is the weight of the initial plastic waste used. The yield of gas was calculated by difference. The reaction time of non-catalytic and catalytic degradation of PE and PP are also presented in Table 2.

As it can be seen, condensable products (Liquid/wax) were by far the major fraction of products with different feedstock and for both processes. Non-catalytic pyrolysis of PE lead to a high yield of wax with 80 wt.% and gas with 20 wt.% and a reaction time of 120 min. On the other hand, the degradation of PP gave 85.5 wt.% of liquid and 14.5 wt % of gas with a reaction time of 80 min and no coke deposits were observed. No wax was observed in the non-catalytic degradation of PP, due to the difference in products structure compared to that of PE. The products obtained from the non-catalytic pyrolysis of PP are branched hydrocarbons which have lower melting points compared to linear alkane. While the products obtained from PE pyrolysis contain large proportion of heavy linear hydrocarbons, with a high carbon chain number which could undergo over cracking reactions [22,23].

When USY zeolite was added with the different plastic wastes type, catalytic decomposition yielded a light yellow liquid and no wax was produced. The catalytic pyrolysis of PE resulted in 71 wt.% of liquid, 26.79 wt.% of gas and produced a deposit of coke with 2.21 wt.% and the reaction lasted 50 min. In the catalytic pyrolysis of PP, the oil yield decreased to 82 wt.% with an increase in the gas yield to 16.76 wt.% and a production of 1.24 wt.% of coke and a reaction time of 45 min. The increasing in liquid yield is due to the decomposition of the heavier hydrocarbons into lighter liquid hydrocarbons over USY zeolite catalyst, while the increase in gas production for PP degradation was due to the fact that the heavier components were converted into gas components through secondary cracking reactions. The formation of coke is due to several mechanisms like aromatization and dehydrogenation on the catalyst surface [24].

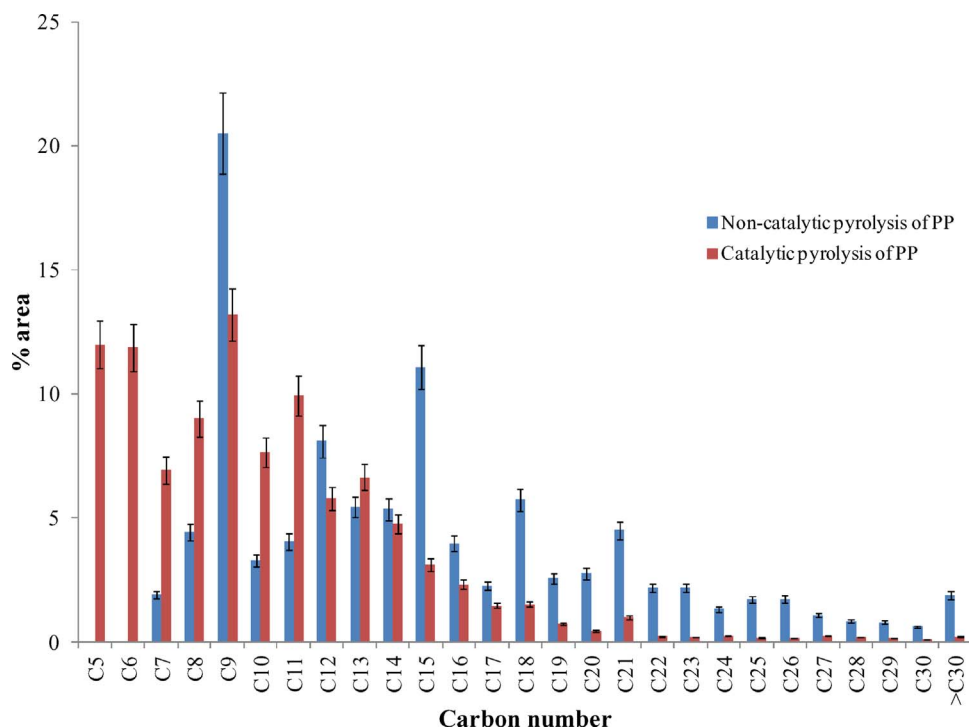


Fig. 2. Carbon number distribution of oil derived from catalytic and non-catalytic pyrolysis of Polypropylene (PP).

### 3.2. Gas chromatography (GC–MS–FID)

The GC–MS FID analysis of the liquid fuel sample obtained by catalytic and non-catalytic pyrolysis of waste PP and PE was carried out to analyze the compounds present in the fuel. Figs. 2 and 3 show the difference in the pyrolysis liquids composition derived from non-catalytic and catalytic pyrolysis of PE and PP respectively, grouped according to their carbon number. As it can be seen in Fig. 2, the liquid obtained from non-catalytic PP pyrolysis contained hydrocarbons ranging from C<sub>7</sub>–C<sub>30</sub> with a maximum peak in C<sub>9</sub> fraction. The use of the catalyst caused a shift of the products into lower molecular weights with products rich in C<sub>5</sub>–C<sub>11</sub> compounds whose yields were as high as 70%. On the other hand, the hydrocarbon composition of liquid oil derived from non – catalytic and catalytic pyrolysis of PE is shown in

Fig. 3. As it can be seen, the liquid hydrocarbon products derived from the non-catalytic pyrolysis were distributed in a wide range carbon numbers (C<sub>8</sub>–C<sub>39</sub>) with a maximum fraction (C<sub>18</sub>–C<sub>21</sub>), while the liquid fraction derived from the catalytic pyrolysis was a mixture of C<sub>5</sub>–C<sub>39</sub> compounds with a maximum fraction (C<sub>10</sub>–C<sub>13</sub>). This changing in composition between non-catalytic and catalytic degradation can be explained by the mechanisms that took place during the process. Thermal degradation of polyolefins has been reported to take place through two mechanisms: (1) random scission to yield long-chain hydrocarbons; and (2) chain-end scission of the oligomers to yield low-molecular weight products [25]. Following these two mechanisms, free hydrogen radicals are formed from the scission of the polymer chain. Thereafter, a molecule with an unsaturated end and the other with a terminal free radical are formed through  $\beta$ -scission. Then, stabilization

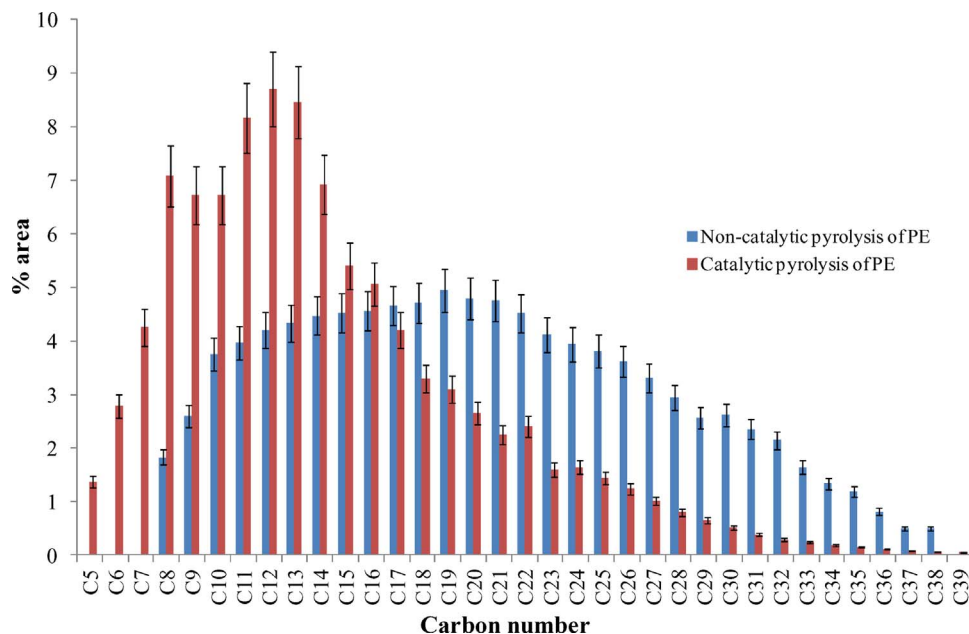


Fig. 3. Carbon number distribution of oil derived from catalytic and non-catalytic pyrolysis of Polyethylene (PE).



**Table 3**

Characteristics of liquid products obtained from non-catalytic and catalytic pyrolysis of PE and PP compared to the European standards.

|                                  | Non-Catalytic Degradation |      | Catalytic Degradation |      | EN 590  | EN 228  |
|----------------------------------|---------------------------|------|-----------------------|------|---------|---------|
|                                  | PE                        | PP   | PE                    | PP   |         |         |
| Plastic type                     | PE                        | PP   | PE                    | PP   |         |         |
| Density (kg/m <sup>3</sup> )     | –                         | 777  | 749                   | 754  | 820–845 | 720–775 |
| Viscosity (mm <sup>2</sup> /s)   | –                         | 0.6  | 0.7                   | 0.6  | 2–4.5   | –       |
| Lower heating value (LHV)(MJ/kg) | –                         | 44.9 | 45.1                  | 45.8 | 43.4    | 44.4    |

of the free radicals occurs by hydrogen chain transfer reactions including intermolecular transfer reaction and intra-molecular transfer reaction. The radical fragments were transformed into straight chain dienes, alkenes and alkanes. Termination reaction occurs by disproportionation of the free radicals or combining of two free radicals [26]. The formation of wax through non-catalytic pyrolysis of PE is related to the fact that PE consists of straight long carbon chains and thermal degradation occurs by random-chain scission. Hydrocarbons are decomposed through the homolytic rupture of covalent C–C bonds, rather than C–H bonds. In this case, a relatively small proportion of branched molecules are detected which accurate a homolytic mechanism.

We can notice that higher volatile compounds were obtained from PP in contrast to PE, using the same pyrolysis parameters. This can be due to the difference in their structure since PP is a polymer whose chains present a greater cross-sectional area than PE, due to the presence of the methyl group substituent on alternate carbon atoms of the chain. The presence of these methyl groups increase the number of tertiary carbons in the molecule, which have considerably lower resistance against degradation and promotes the thermal cleavage of C–C bonds and due to the branched structure, methyl, primary and secondary alkyl radicals are formed [27]. In fact, the mechanism of the thermal degradation of polypropylene and polyethylene initiates by random chain scission with a formation of secondary and tertiary radicals via rearrangement reactions (backbiting) and propagates through the  $\beta$ -scission of the radicals [28]. The difference in the cracking mechanism between PP and PE and the formation of secondary and tertiary radicals was explained by the lower degradation reaction time of PP compared to that of PE. Transfer reactions play a minor role in polypropylene degradation compared to polyethylene degradation due to the lower alkane concentration [29]. The specific carbon number distribution presented in Fig. 2 with maximum at C<sub>9</sub>, C<sub>12</sub>, C<sub>15</sub>, C<sub>18</sub> and C<sub>21</sub> showed preferential cracking position, corresponding to the tertiary carbons.

To explain the products observed from the catalytic cracking of PE and PP over USY zeolite, two carbocationic mechanisms have been suggested: (1) the classical bimolecular mechanism or  $\beta$ -cracking [30] and (2) the monomolecular mechanism, through carbonium ions [31]. Catalytic cracking of PE initiates with formation of free radicals along the polymer chains, which results in breakage of C–C bond. Due to abstraction of hydride ion from the PE chain by USY, carbenium ion is formed and undergoes isomerization,  $\beta$ -scission, and hydrogen transfer to form smaller fragments [32]. Catalytic degradation of PP is known to proceed by carbenium ion mechanism [33]. The catalytic degradation of PP initiates by either the abstraction of hydride ion (by Lewis acid sites) from the oligomer chains, or the attack of one Brønsted acid site to the double bonds of the thermally degraded PP molecules. Fragments having lower molecular weights were produced in the catalytic degradation of PP compared to that of PE, since the catalyst has a higher catalytic cycle in order to degrade the PE chain than that of PP, and this is consistent with the higher degradation reaction time of PE than that of PP. It is well known that the pyrolysis of PE and PP produce olefins and paraffin but the use of USY catalyst modified their composition while it is difficult to notice a variation in PP pyrolysis product, PE pyrolysis product were radically changed. There are no normal paraffins recovered in catalytic condition while in non catalytic cracking

they represent more than 70% of the recovered oil. Olefins have much higher reactivity over acidic catalysts than paraffins since they can be easily converted to carbocations on zeolite surface.

### 3.3. Physical properties of oil sample

From the Gas chromatograph analysis, there was a significant difference among molecular weights of oils obtained by pyrolysis of PE and PP in the presence and absence of catalyst.

The main properties of the pyrolysis oils obtained from the non-catalytic and catalytic pyrolysis of PP and PE at 450 °C are summarized in Table 3. The non-catalytic pyrolysis of PE gave a wax with a high viscosity, while the non-catalytic pyrolysis of PP gave a less viscous liquid. Those properties are directly linked to the composition of those oils. Non-catalytic pyrolysis oil from PE is essentially composed of normal paraffins/olefins with high viscosity while pyrolysis oil from PP is a mixture of branched paraffins/olefins with low viscosity. In non-catalytic conditions, reaction time was too long and consumed a lot of energy, and produced a liquid which could not be recovered into valuable fuel oil. For this reason, the rest of the present study will be focused on the catalytic pyrolysis of both PP and PE.

As it can be seen, liquids obtained by catalytic pyrolysis of waste PP and PE have LHVs in the range of gasoline and diesel. According to ASTM D 975, there are no limits for density of diesel fuel, but EN 590 specifies a density of 820–845 kg/m<sup>3</sup> at 15 °C. Neither the liquid oil obtained from pyrolysis of PP (754 kg/m<sup>3</sup>) nor the liquid oil obtained from pyrolysis of PE (749 kg/m<sup>3</sup>) provided densities meeting the range specified in EN 590. On the other hand, both liquids densities meet the range of gasoline specified in EN 228.

Viscosities of both pyrolytic oils obtained from pyrolysis of PP (0.64 mm<sup>2</sup>/s) and from pyrolysis of PE (0.72 mm<sup>2</sup>/s), were out of the EN590 range and are very low compared to conventional diesel fuel (2–4.5 mm<sup>2</sup>/s). This could be explained by the presence of light olefins and the presence of high branched isomers due to the carbon-chain isomerization that takes place in the presence of catalysts [34].

**Table 4**

Distillation properties of the produced oils according to EN 228 standard limits established for gasoline and EN 590 standard limits established for diesel.

| Distillation Properties     | Units   | limits EN228 | Feedstock |      |
|-----------------------------|---------|--------------|-----------|------|
|                             |         |              | PE        | PP   |
| Evaporated @ 70 °C          | % (V/V) | 20–48        | 10        | 9    |
| Evaporated @ 100 °C         | % (V/V) | 46–71        | 27        | 26   |
| Evaporated @ 150 °C         | % (V/V) | 75(min)      | 49        | 47   |
| Initial boiling point (IBP) | °C      | –            | 46.8      | 47.7 |
| Final boiling point (FBP)   | °C      | 210 (max)    | 352       | 346  |
| Distillation Properties     | Units   | limits EN590 | Feedstock |      |
|                             |         |              | PE        | PP   |
| Recovered @ 250 °C          | % (V/V) | 65 (max)     | 76        | 77   |
| Recovered @ 350 °C          | % (V/V) | 85 (min)     | 89        | 91   |
| Temp @ 95% V/V rec          | °C      | 360 (max)    | 360       | 360  |

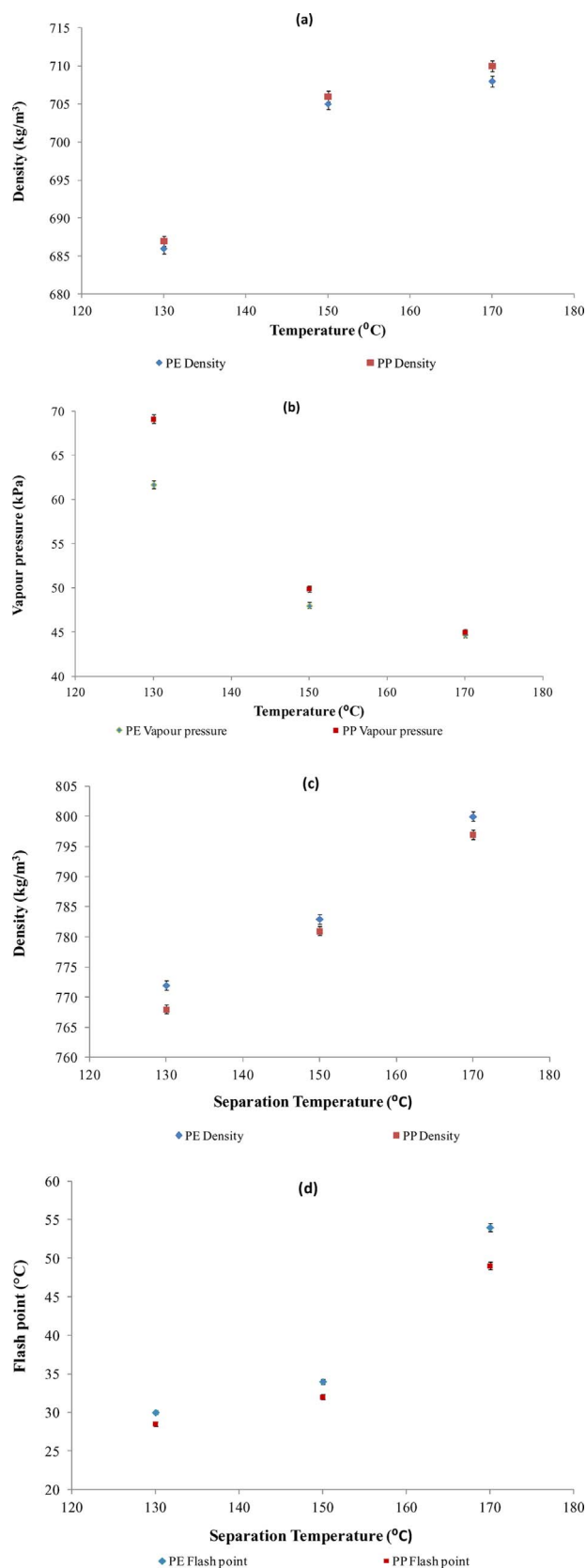


Fig. 4. Physical properties of (a,b) light phase and (c,d) heavy phase at different separation temperatures.

### 3.4. Distillation fraction of pyrolytic oils

Table 4 shows the boiling range distributions of the oil obtained from the catalytic pyrolysis of PP and PE in the presence of USY zeolite compared to the two major commercial fuels, gasoline and diesel. It is observed that the boiling range of the oil is out of the range of gasoline and diesel because of the presence of a mixture of light and heavy compounds, which suggests the separation of oils into different fractions as gasoline-like, kerosene-like and diesel-like fuels. According to EN228 requirements, minimum 20% of the volume should be evaporated at 70 °C, and in order to meet this requirement, the max final boiling point was 170 °C. The oil obtained from the pyrolysis was after separated by distillation to two fractions, a heavy phase and a light phase, at 3 different temperatures (130, 150 and 170 °C). The fuel properties of the different fractions are presented in Fig. 4.

As it can be seen in Fig. 4 (a), the optimal temperature of separation is 170 °C where the densities of the light phase from the liquid oil obtained from pyrolysis of PP and the liquid oil obtained from pyrolysis of PE meet the range of gasoline specified in EN 228. In addition, according to the volatility classes, the light phase ranged in classes A & B has a vapor pressure between 45 and 70 kPa. As shown in Fig. 4 (b), at a separation temperature of 170 °C, the light phase belongs to classes A & B.

Fig. 4 (c, d) shows the densities and the flash point of the heavy phase obtained after distillation of the pyrolytic oils at different temperatures. The densities and flash point obtained at 170 °C are very close to the range of EN 590.

Tables 5 and 6 show the boiling range distributions of the oil after separation at 170 °C compared to the two major commercial fuels,

Table 5

Distillation and physical properties of the gasoline-like oil obtained after separation @ 170 °C according to EN 228 standard limits established for gasoline fuel.

| Distillation Properties | Units             | limits EN228    | Feedstock |      |
|-------------------------|-------------------|-----------------|-----------|------|
|                         |                   |                 | PE        | PP   |
| IBP                     | °C                | –               | 46.8      | 47.7 |
| Evaporated @ 70 °C      | % (V/V)           | 20–48           | 21        | 21   |
| Evaporated @ 100 °C     | % (V/V)           | 46–71           | 52        | 50   |
| Evaporated @ 150 °C     | % (V/V)           | 75              | 90        | 87   |
| FBP                     | °C                | 210 (max)       | 170       | 170  |
| Physical Properties     |                   |                 |           |      |
| RON                     |                   | 95 (min)        | 97        | 96   |
| Density @ 15 °C         | kg/m <sup>3</sup> | 720–775         | 708       | 710  |
| Vapour pressure         | kPa               | 45–60 (class A) | 44.7      | 45   |

Table 6

Distillation and physical properties of the diesel-like oil obtained after separation @ 170 °C according to EN 590 standard limits established for diesel fuel.

| Distillation Properties  | Units              | limits EN590 | Feedstock |       |
|--------------------------|--------------------|--------------|-----------|-------|
|                          |                    |              | PE        | PP    |
| IBP                      | °C                 | –            | 159.5     | 160.1 |
| Recovered @ 250 °C       | % (V/V)            | 65 (max)     | 61        | 60    |
| Recovered @ 350 °C       | % (V/V)            | 85 (min)     | 100       | 100   |
| Temp @ 95% V/V rec       | °C                 | 360 (max)    | 350       | 350   |
| Physical Properties      |                    |              |           |       |
| Calculated cetane number |                    | 46 (min)     | 53        | 52    |
| Density @ 15 °C          | kg/m <sup>3</sup>  | 820–845      | 797       | 800   |
| Viscosity @ 40 °C        | mm <sup>2</sup> /s | 2–4.5        | 1.4       | 1.5   |
| Flash point              | °C                 | 55 (min)     | 54        | 49    |

gasoline and diesel. As it can be seen, at 170 °C, the boiling points of fractions fit completely those of diesel and gasoline and their characteristics are very close to those of gasoline and diesel. For that reason, the octane and cetane number were calculated and viscosities of the liquid oil obtained after distillation at 170 °C were measured. The viscosity has increased with separation temperature until reaching 1.4 mm<sup>2</sup>/s (PE) and 1.5 mm<sup>2</sup>/s (PP). The viscosity is still lower than that of conventional diesel fuel (2–4.5 mm<sup>2</sup>/s). The light phase which represents 60.6% of the total liquid of PP and 57% of that of PE had respectively a high octane number (RON=96 and RON=97), while the heavy phase representing 36.5% of the total liquid of PP and 35.3% of that of PE had a calculated cetane number of 52 (PP) and 53 (PE) which meets the EN 590 requirements. In order to improve the physical properties of diesel-like fuel, blending it with commercial diesel fuel could be considered.

#### 4. Conclusions

In this work, non-catalytic and catalytic pyrolysis of polyethylene and polypropylene has been carried out in a batch reactor at 450 °C. Non-catalytic pyrolysis of PE produced 80 wt.% of wax and 20 wt.% of gas. On the other hand, the degradation of PP gave 85.5 wt.% of liquid and 14.5 wt.% of gas. When USY catalyst was added, pyrolysis of PE produced 71 wt.% of liquid and 26.79 wt.% of gas and produced a deposit of coke with 2.21 wt.% while in the catalytic pyrolysis of PP, the oil yield decreased to 82 wt.% with an increase in the gas yield to 16.76 wt.% and a production of 1.24 wt.% of coke.

Based on the GC analysis, the liquid obtained from non-catalytic pyrolysis of PP contained branched hydrocarbons with a maximum carbon number distribution at C<sub>9</sub>, C<sub>12</sub>, C<sub>15</sub>, C<sub>18</sub> and C<sub>21</sub> showing a preferential cracking position, while the liquid obtained from non-catalytic pyrolysis of PE contained normal hydrocarbons with a maximum fraction ranging from C<sub>18</sub>–C<sub>21</sub>. The use of the catalyst caused a shift of the products into lower branched molecular weights. The oil derived from catalytic pyrolysis of PP contained products rich in C<sub>5</sub>–C<sub>11</sub> compounds whose yields were as high as 70% while the liquid fraction derived from the catalytic pyrolysis of PE was rich in C<sub>10</sub>–C<sub>13</sub> compounds.

Compared to conventional fuels, Fractionation at 170 °C of PE and PP pyrolytic oils resulted in light and heavy fractions can meet the norms of gasoline and diesel fuels, respectively.

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